Yangze Liu

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Education

University of Illinois Urbana-Champaign Aug 2021 – May 2025
B.S. in Liberal Arts and Sciences, Computer Science + Philosophy
GPA: 3.77/4.0

Research Interests

I study LLM-based agents, using controlled experiments to understand what actually drives their behavior and performance: where the knowledge in an agent's memory comes from, whether a model is coupled to its harness, whether a benchmark rewards the decisions it claims to test, and what goes wrong in an agent's actions even when it passes its task. I want to use this understanding to build better agents that can run safely in real environments. Earlier I worked on data selection for efficient training.

Publications

* Equal contribution.

Published and accepted

[1] Yibing Liu*, **Yangze Liu***, Xiaolong Yin, Bin Wang, Chong Zhang, Hao Yin, and Zhongyi Han. OpenClawBench: Benchmarking Process-side Anomalies in Real-world Agent Execution Trajectories. *Advances in Neural Information Processing Systems (NeurIPS), Evaluations & Datasets Track*, 2026. Poster. arXiv:2605.29253.

[2] **Yangze Liu** and Hong Liu. Rethinking Dataset Quantization: Efficient Coreset Selection via Semantically-Aware Data Augmentation. *Transactions on Machine Learning Research (TMLR)*, 2026. OpenReview.

Under review: LLM agents

[3] **Yangze Liu** and Zhongyi Han. What Does a Harness Buy? Tokens, Mostly. Under review at ICLR 2027. arXiv:2610.04433.

[4] **Yangze Liu** and Zhongyi Han. Experience Is Not Knowledge: Where the Decision Knowledge in an Agent's Memory Comes From. Under review at ICLR 2027.

[5] **Yangze Liu** and Zhongyi Han. Asking Earns Nothing: Scoring the Decision to Act in BFCL Multi-Turn. Under review at ICLR 2027. arXiv:2610.04429.

Research Experience

Shandong University Oct 2025 – Present
Full-time Research Assistant, advised by Prof. Zhongyi Han

- **OpenClawBench** (co-first author) [1]. A benchmark of process-side anomalies in real agent execution trajectories, where an agent passes the final task check but still ignores errors, makes unsafe external writes, or commits to weakly grounded claims. Built the 31,264-trajectory dataset from generation through cleaning, and trained its trajectory-level anomaly detector, a LoRA-fine-tuned open model.
- **Is a model coupled to its harness?** [3] Coding-agent harnesses ship new releases constantly and are credited with pass-rate gains, but whether the same model actually does better in a different harness had rarely been separated from run-to-run noise. On a hard subset of SWE-bench Verified, comparisons across five models showed that harness swaps flipped as many task outcomes as same-configuration reruns. On the larger task pool, tested with two models, the only harness effect that cleared the noise was a loss. What the harness did decide was cost: up to three times per task for the same model.
- **Where the knowledge in an agent's memory comes from.** [4] Agents are often improved by writing a memory from their own trajectories, and the gain is credited to experience, but such a memory mixes what the writer already knew with what it actually learned. Built a controlled Slay the Spire testbed that varies only the source of the memory, and designed the memory-construction methods it compares. The agent's self-written memory showed no detectable improvement in clear rate. Simulator-statistics and stronger-model-prior memories had higher observed clear rates than the no-memory baseline. The teacher-memory gain survived multiple-comparison correction. Replay at shared decision states showed that the agent acts on its own memory, and that a wrong rule it drew from its own games became a new error.
- **Does a benchmark reward the decision to ask?** [5] An agent that lacks the information it needs should ask rather than act, and BFCL multi-turn builds two of its four categories around that decision, yet its scorer skips the very turn on which the agent should ask, so asking earns no credit. Built a judge-free decision metric from the

benchmark's own paired complete and incomplete requests. A one-line prompt pushing an agent toward action raised official scores on the two should-ask categories and their base twins by 13.5–23.5 percentage points with no detectable change in its decision accuracy, and the model that decided best ranked sixth of seven tested models under official scoring on the same evaluated items. Also flagged 31 defective items by hand, 22 with wrong answer keys.

- **Auditing public agent leaderboards.** Audited HAL, τ-bench, and τ^2-bench as measurement instruments and reported four reproducible scoring and cost-accounting faults to their maintainers with reproduction scripts. In one, an agent on τ-bench Airline was graded in only 39% of episodes in the auditable subset, and regrading all episodes in that subset reordered the scaffolds for six of ten models covered by at least two scaffolds. The HAL maintainers reproduced and confirmed three of the faults and added disclosures to the leaderboard.

Research collaboration with Prof. Hong Liu Jan 2024 – Jun 2026

Prof. Liu was then at Osaka University and is now at Xiamen University

Dataset Quantization relies on a large pretrained model and augments data only after selecting it, so selection and training see different data. Developed a version that finds objects for semantically aware augmentation with a randomly initialized CNN and augments before selecting. It is competitive with or better than prior methods on ImageNet-1k and fine-grained datasets, at lower cost than the original method; first-author paper in TMLR [2].

Other Manuscripts

All under review at ICLR 2027.

Data selection

[6] Are Coreset Selection Methods Worth Their Cost? **Y. Liu**, Z. Han. arXiv:2609.22894.

[7] When Hard-Example Selection Fails: Gaussian Thresholds and Empirical Diagnostics. **Y. Liu**, Z. Han.

[8] What Does 5% Mean? The Coreset Regime Boundary Is Pinned in Samples per Class. **Y. Liu**, X.-L. Yin, Z. Han.

[9] Useful to Whom? The Easy-versus-Coverage Boundary Moves with the Learner. **Y. Liu**, X.-L. Yin, Z. Han. arXiv:2610.00221.

Recursive training

[10] A Dominant Supplier Slows Recursive Drift More Than It Steers It. **Y. Liu**, Z. Han. arXiv:2609.11146.

[11] Break Step: Recursive Training Resonates with Replayed Sampling Noise. **Y. Liu**, Z. Han. arXiv:2609.11149.

Other Experience

University of Illinois Urbana-Champaign Oct 2023 – May 2025

Research Assistant, advised by Prof. Nishant Garg

Led development of Crystal Vision, a cross-platform VR and PC educational game built in Unreal Engine 5 and adopted in CEE 300 (Behavior of Materials) to teach crystallography. The project later received research funding for continued development.

Service

Reviewer, International Conference on Learning Representations (ICLR) 2027

Skills

Programming	Python, C/C++, Java
ML and LLM systems	PyTorch, vLLM (tensor-parallel inference serving), LoRA fine-tuning
Agents and benchmarks	Claude Code, mini-SWE-agent, OpenCode, Harbor; SWE-bench, BFCL, τ-bench, GAIA
Infrastructure	Linux, multi-GPU experiment management across shared servers
Languages	Mandarin Chinese, English

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